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Laporan Pratikum 1 Cyber Physical System

Verifikasi dilakukan dengan memeriksa berkas konfigurasi Docker, package ROS 2, proses komunikasi antar-node, serta riwayat commit pada repositori Git. Berkas konfigurasi yang tersedia meliputi docker/Dockerfile, docker/entrypoint.sh, dan docker/compose.yaml. Selain itu, package kustom my_first_robot_package telah dibuat dan memiliki struktur package ROS 2 yang diperlukan.

```
nay@Pongo:~/robotics-turtlebot3$ find docker -maxdepth 1 -type f -print
docker/compose.yaml
docker/entrypoint.sh
docker/Dockerfile
docker/commpose.yaml
nay@Pongo:~/robotics-turtlebot3$
```

```
nay@Pongo:~/robotics-turtlebot3$ ls src/my_first_robot_package
my_first_robot_package  package.xml  resource  setup.cfg  setup.py  test
nay@Pongo:~/robotics-turtlebot3$
```

Pengujian menggunakan Docker Compose berhasil menjalankan dua container, yaitu m01_talker dan m01_listener. Output menunjukkan bahwa node /talker berhasil melakukan Publishing pesan Hello World, sedangkan /listener berhasil menerima pesan tersebut dengan output “I heard: [Hello World: ...]”

```
nay@Pongo:~/robotics-turtlebot3/docker$ docker compose up
[+] up 2/2
 ✔Container m01_talker    Created                                          0.5s
 ✔Container m01_listener  Created                                          0.2s
Attaching to m01_listener, m01_talker
m01_talker    | [INFO] [1790775151.085773637] [talker]: Publishing: 'Hello World: 1'
m01_listener  | [INFO] [1790775151.110291739] [listener]: I heard: [Hello World: 1]
m01_talker    | [INFO] [1790775152.085900450] [talker]: Publishing: 'Hello World: 2'
m01_listener  | [INFO] [1790775152.088557421] [listener]: I heard: [Hello World: 2]
m01_talker    | [INFO] [1790775153.085954184] [talker]: Publishing: 'Hello World: 3'
m01_listener  | [INFO] [1790775153.087393784] [listener]: I heard: [Hello World: 3]
m01_talker    | [INFO] [1790775154.086114428] [talker]: Publishing: 'Hello World: 4'
m01_listener  | [INFO] [1790775154.089164293] [listener]: I heard: [Hello World: 4]
m01_talker    | [INFO] [1790775155.086059356] [talker]: Publishing: 'Hello World: 5'
m01_listener  | [INFO] [1790775155.087415067] [listener]: I heard: [Hello World: 5]
m01_talker    | [INFO] [1790775156.086173518] [talker]: Publishing: 'Hello World: 6'
m01_listener  | [INFO] [1790775156.088754325] [listener]: I heard: [Hello World: 6]
m01_talker    | [INFO] [1790775157.086180997] [talker]: Publishing: 'Hello World: 7'
m01_listener  | [INFO] [1790775157.087625493] [listener]: I heard: [Hello World: 7]
m01_talker    | [INFO] [1790775158.086299802] [talker]: Publishing: 'Hello World: 8'
m01_listener  | [INFO] [1790775158.087860444] [listener]: I heard: [Hello World: 8]
```

Pemeriksaan ros2 node list juga menunjukkan dua node yang aktif, yaitu /talker dan /listener. Pengujian ros2 topic echo /chatter --once berhasil menampilkan pesan data: Hello World: 169, sehingga komunikasi melalui topic /chatter dapat dibuktikan berjalan dengan baik.

```
nay@Pongo:~/robotics-turtlebot3/docker$ docker compose run --rm talker ros2 node list
Container docker-talker-run-0f7c8c74d74a Creating
Container docker-talker-run-0f7c8c74d74a Created
/listener
/talker
nay@Pongo:~/robotics-turtlebot3/docker$ docker compose run --rm talker ros2 topic echo /chatter
 --once
Container docker-talker-run-4610e7f7832e Creating
Container docker-talker-run-4610e7f7832e Created
data: 'Hello World: 169'
---
```

Repositori Git juga telah memiliki beberapa commit bermakna, antara lain commit untuk dokumentasi laporan, README, dan konfigurasi environment ROS 2. Hal tersebut menjadi bukti bahwa proses pengerjaan dan dokumentasi proyek telah tercatat dalam repositori.

```
   bf8c841..46788e1  main -> main
nay@Pongo:~/robotics-turtlebot3$ git log --oneline
46788e1 (HEAD -> main, origin/main) docs: add module 01 report
bf8c841 docs: add README
5d09906 feat(m01): dockerized ROS 2 Humble dev environment
nay@Pongo:~/robotics-turtlebot3$
```